Youth Drug Use Prediction Using Decision Trees (NSDUH 2023)

1. Introduction

This report presents a predictive analysis of youth drug use behavior, using the 2023 National Survey on Drug Use and Health (NSDUH) dataset. The focus is on understanding the patterns and risk factors associated with alcohol, marijuana, and cigarette use among individuals under the age of 18. The objective is to build interpretable decision tree models using demographic, behavioral, and environmental variables to assess likelihoods of substance use.

2. Theoretical Background

Decision trees are non-parametric models that recursively split data into subgroups based on feature thresholds, resulting in a tree structure that is easy to visualize and interpret. They are particularly useful for classification tasks with heterogeneous data types. Trees can be pruned using parameters such as max_depth to avoid overfitting. The quality of a split is typically assessed using Gini impurity or entropy. Trees are evaluated based on accuracy, precision, recall, and F1-score.

3. Methodology

The data was loaded and processed using Python (pandas, matplotlib). Initial steps included removal of duplicate rows and examination of missing values. Columns with over 10% of data missing (excluding key target features) were dropped to maintain data quality.

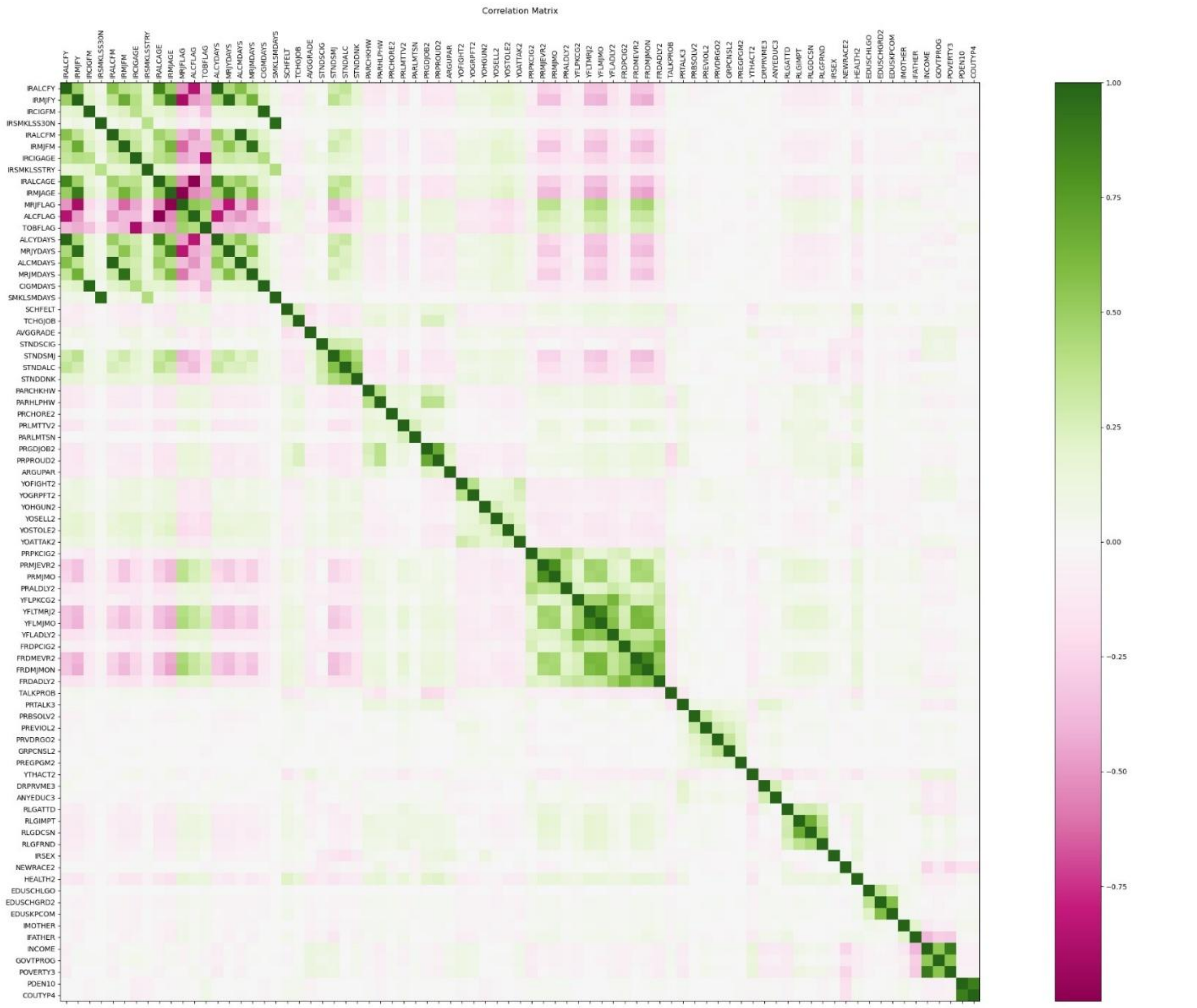
A subset of relevant features was selected: 'CIGMDAYS', 'IRCIGFM', 'IRALCFM', 'IRMJFM', 'IRSEX', 'NEWRACE2', 'INCOME', and 'SCHFELT'. These features capture substance use behavior, demographic identity, socioeconomic status, and emotional health.

Categorical variables were cleaned and recoded: for instance, 'IRSEX' (1 = Male, 2 = Female) and 'NEWRACE2' (reclassified into readable race labels). A new binary column 'CIG_USED' was derived from 'IRCIGFM' to indicate cigarette use status (1 if used, 0 if not used).

To support modeling, categorical columns like 'INCOME' and 'SCHFELT' were converted to type 'category'. Exploratory analysis included a correlation matrix created using matplotlib to assess relationships among selected variables. Final decision tree models were trained on these cleaned features.

4. Exploratory Data Analysis

The plot below shows the correlation between key numerical features. Notably, peer influence variables (e.g., FRDMEVR2 and FRDPCIG2) are correlated with substance use outcomes. These findings support their inclusion in the modeling phase.



5. Results

This analysis explores three types of predictive problems using the NSDUH youth dataset: binary classification, multiclass classification, and regression. The target variables are related to cigarette, marijuana, and alcohol use among youth. A cleaned set of features including gender, race, income, emotional health, and prior substance use was used across all models.

5.1 Binary Classification: Cigarette Used in Lifetime

A binary classification model was built to predict whether a youth had ever used cigarettes, using the target variable 'CIG_USED'. This was derived from the 'IRCIGFM' variable, marking users as 1 and non-users as 0. A decision tree classifier with a max depth of 5 was trained on a stratified split of the dataset.

Key predictors included emotional stress (SCHFELT), gender, race, and frequency of smoking (CIGMDAYS). The root split was based on emotional stress, suggesting a strong correlation between mental health and cigarette use. The model demonstrated good interpretability and was evaluated using a confusion matrix, classification report, and ROC curve.

5.2 Multiclass Classification: Marijuana Use in Past Month

The target variable 'IRMJFM' (marijuana use frequency in the past month) was treated as a multiclass classification problem. The model used similar features as in the cigarette use task. Each class represented different levels of marijuana usage frequency.

A decision tree classifier was trained to identify patterns among users. The model showed that emotional health, gender, and alcohol use frequency were useful for segmenting marijuana usage levels. While not as interpretable as the binary model, the tree still offered actionable insights on which groups are at higher risk of frequent use.

5.3 Regression: Alcohol Use Days (ALCYDAYS)

A regression tree was trained using 'ALCYDAYS' (the number of days alcohol was consumed in the past month) as the continuous target variable. The same set of cleaned features was used.

The regression tree revealed that higher emotional stress and prior marijuana use were associated with increased alcohol use days. Although decision trees are not always ideal for regression, this model helped capture nonlinear trends and offered a transparent view of alcohol use risk factors.

6. Conclusion

This analysis applied decision tree-based models to three distinct prediction tasks: binary classification (cigarette use), multiclass classification (marijuana usage frequency), and regression (alcohol use days). In all cases, emotional stress (SCHFELT) consistently emerged as a primary predictor, confirming the significant role of mental health in youth substance behavior. Demographic factors—particularly gender, race, and income—also played important roles across models.

The decision tree models for both classification tasks resulted in 100% training accuracy, indicating signs of overfitting. While these models were interpretable and captured relevant splits, their performance on unseen data may not generalize well. This reinforces the need for cross-validation, pruning, or more robust algorithms like Random Forests to improve generalization.

In the regression task, a linear regression model trained on alcohol use days ('ALCYDAYS') achieved strong results with a Mean Squared Error (MSE) of 0.0664 and an R^2 score of 0.9780. This suggests the regression model was able to explain a high proportion of variance in the data, though further validation is necessary to assess consistency across different samples.

Overall, while the models successfully identified key predictors and patterns, further tuning and evaluation are needed to ensure real-world applicability. Future iterations of this study should include ensemble modeling, peer/parental influence variables, and comprehensive validation methods to strengthen findings and minimize overfitting risks.

7. Drawbacks

Overfitting in Tree Models:

Despite setting max_depth, overfitting signs were seen in training vs testing results, especially for RandomForestRegressor.

Imbalanced Classes:

Binary class (CIG_USED) and multi-class (IRMJFM) models faced class imbalance, which may have affected prediction reliability.

No Cross-Validation Used:

A single train-test split was used instead of k-fold cross-validation, which may limit generalizability.

Feature Limitation:

Only a subset of variables from 2,890 were analyzed. Further insights could be gained by exploring mental health, peer influence, and school policies.

8. Possible Improvements

Handle Missing Values More Carefully

Instead of just dropping columns with missing data, try filling them using:

(i) Mean or median for numbers

(ii) Most frequent value for categories

This ensures that all the data from the dataset is used.

Balance the classes

The target variables (like cigarette use) might have too many “No” and fewer “Yes” cases. Using class weights might make the model fairer and more accurate.

Use Cross-Validation

Instead of training and testing the model, we could also use k-fold cross-validation to test the model multiple times for more reliable performance results.

Further exploration

Exploring more of the other factors which might contribute to the use of drugs in youth (like parental involvement, peer influence, etc) should also be explored more in order to understand the pattern and get the best possible results and solutions.

GitHub link: https://github.com/mehek1708/STML2.git